Introduction to Theoretical Computer Science

CS2490 - Session 4: Deterministic Finite Automata

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Deterministic Finite Automata (DFA)

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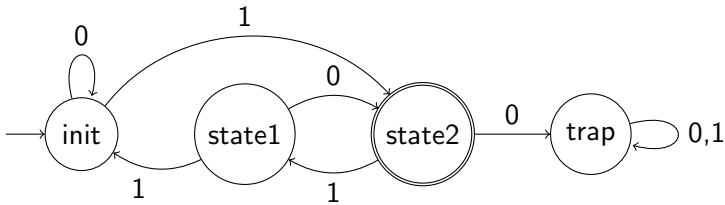
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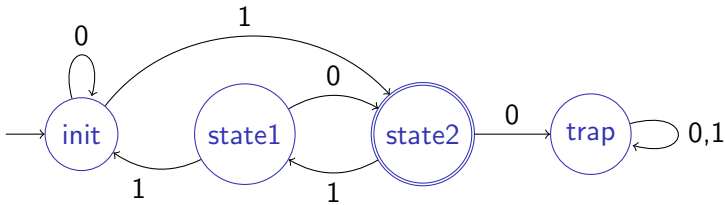
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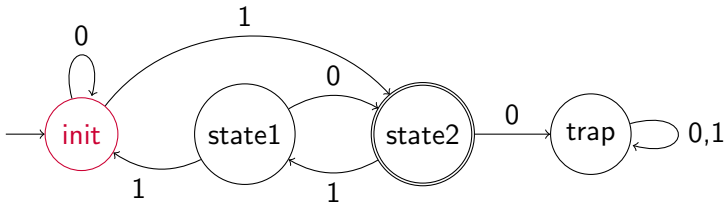
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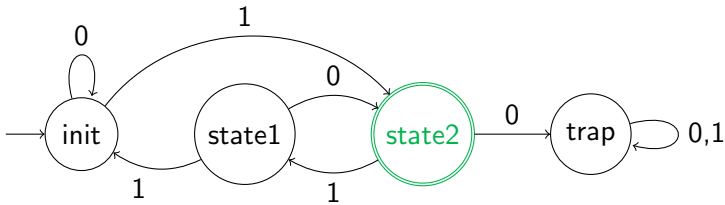
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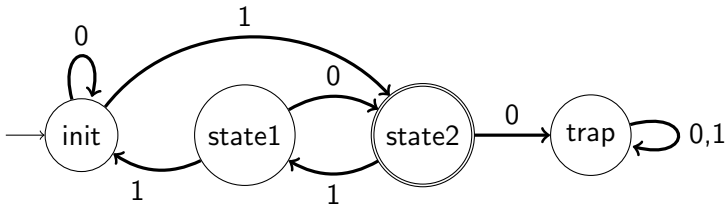
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Computation

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The computation (or run) of $\mathcal{A}$ on a word $w = w_1 w_2 \dots w_n \in \Sigma^*$ is the sequence of states $s_0, s_1, \dots, s_n \in Q$ defined by:

- $s_0 = q_0$,
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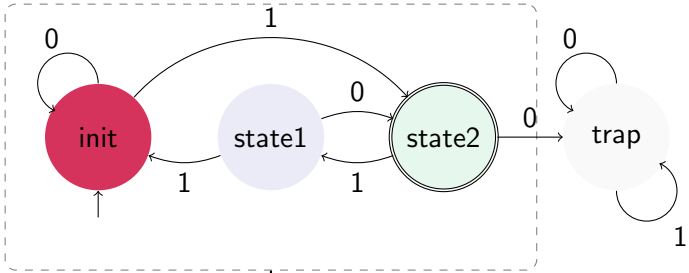
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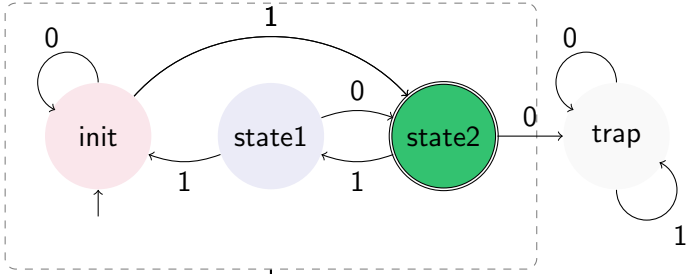
The computation is successful when $s_n \in F$. We then say that $\mathcal{A}$ accepts w.

Example of computation



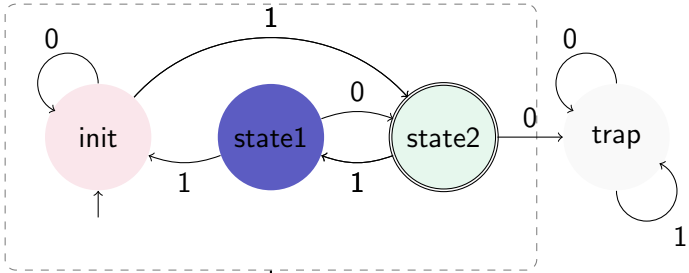
init,

Example of computation



init, state2,

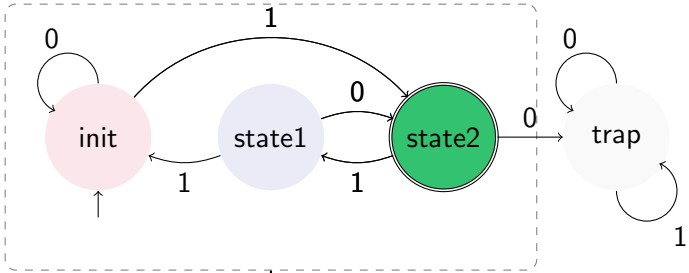
Example of computation



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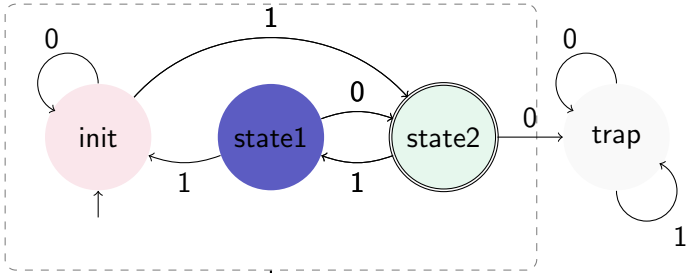
init, state2, state1,

Example of computation



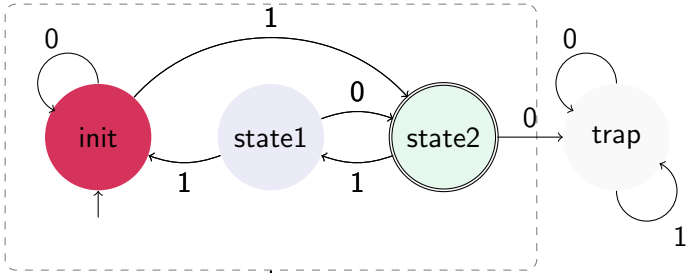
init, state2, state1, state2,

Example of computation



init, state2, state1, state2, state1,

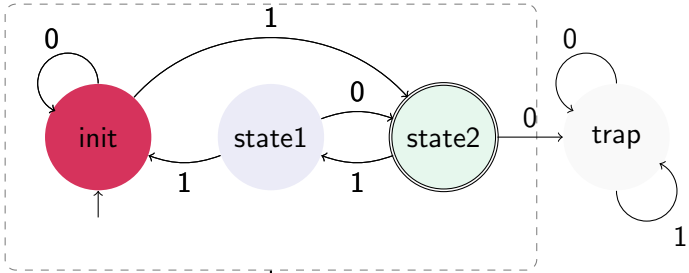
Example of computation



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init, state2, state1, state2, state1, init,

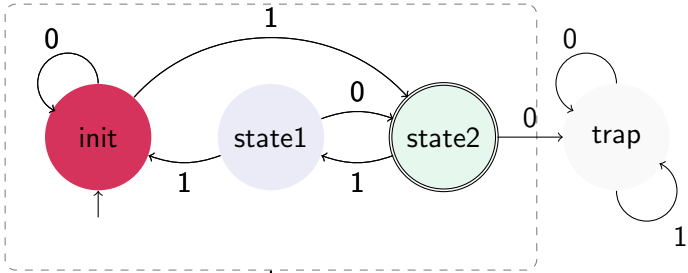
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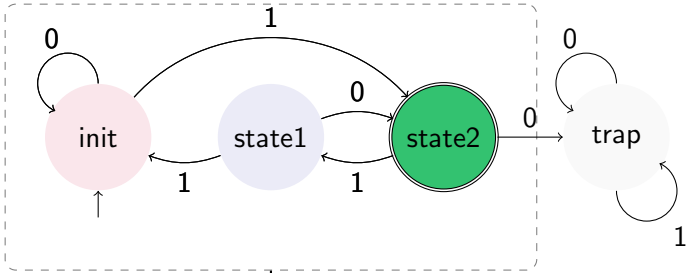
init, state2, state1, state2, state1, init, init,

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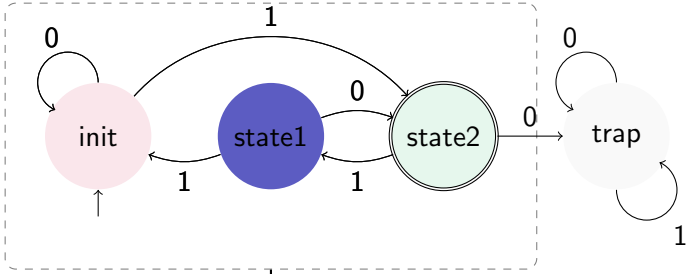
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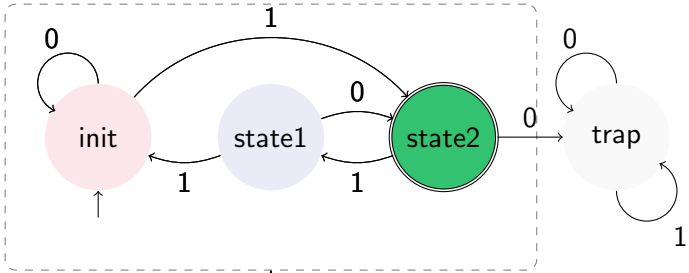
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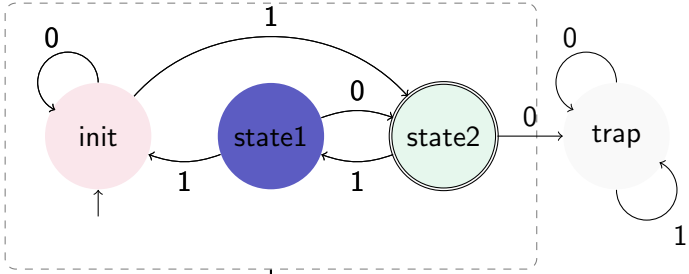
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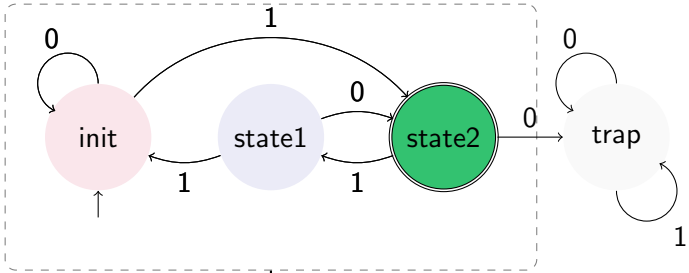
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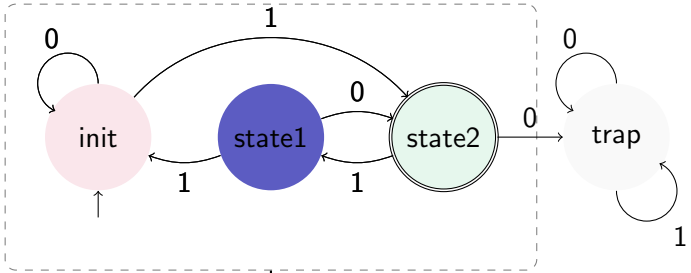
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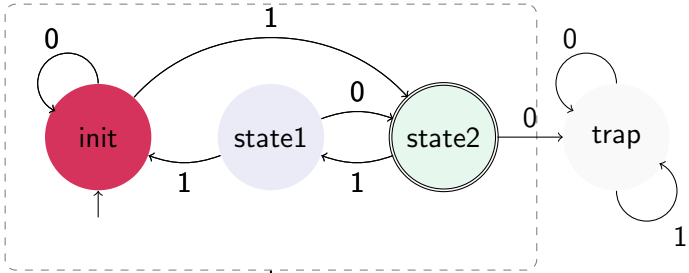
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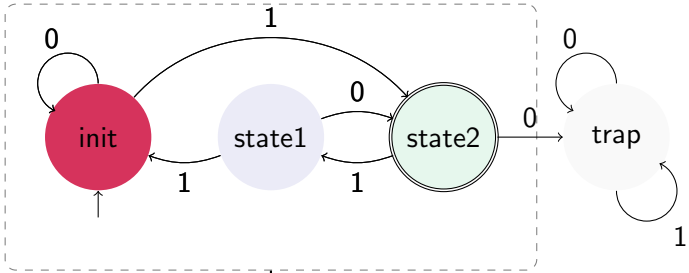
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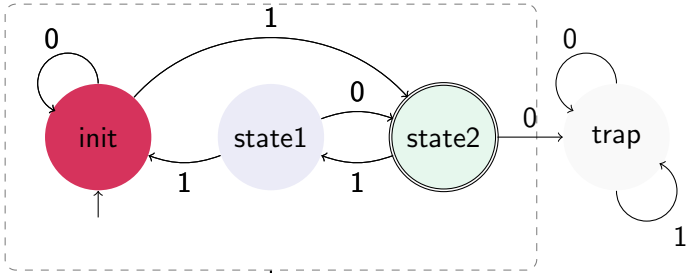
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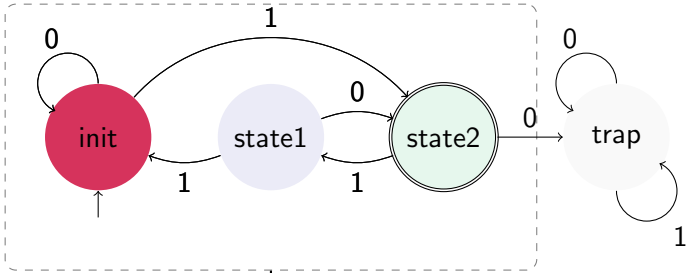
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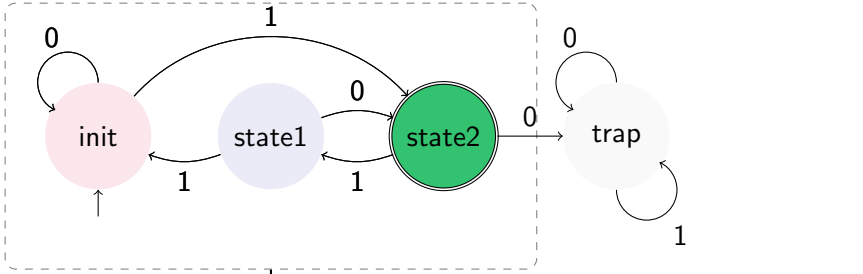
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In other words, the extended transition function $\delta^* : \Sigma^* \times Q \to Q$ is defined as follows:

- $\delta^*(\varepsilon, q) = q$ for all $q \in Q$.
- $\delta^*(wa, q) = \delta(a, \delta^*(w, q))$ for all $a \in \Sigma$, $w \in \Sigma^*$, and $q \in Q$.

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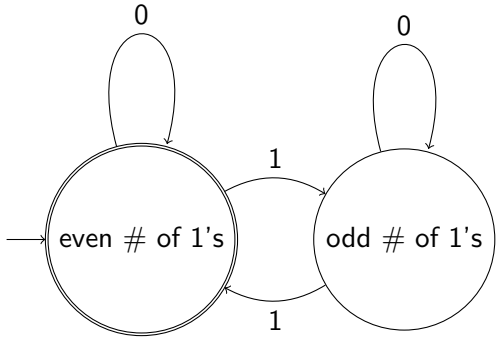
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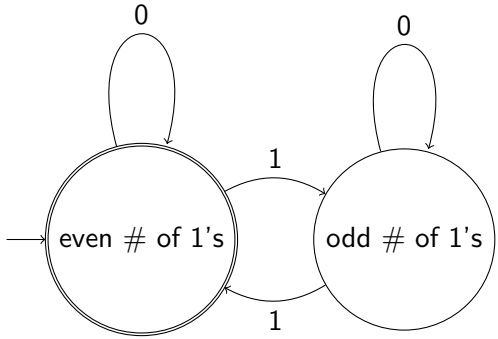
$$\mathrm{L}(\mathcal{A}) = \{w \in \Sigma^* \mid \delta^*(w, q_0) \in F\}.$$

$\mathrm{L}(\mathcal{A})$ is also called the language recognized by $\mathcal{A}$, or even more simply the language of $\mathcal{A}$.

Example

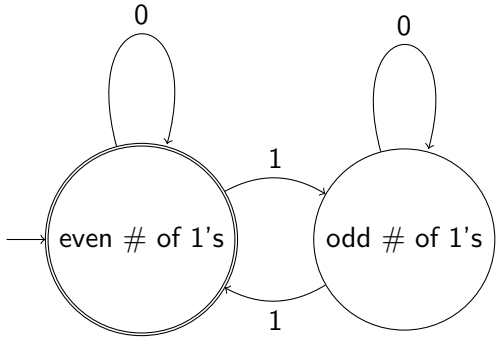


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Guess: $L(\mathcal{A}) = ?$

Example



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How to prove that $\mathrm{L}(\mathcal{A}) = \{w \in \{0,1\}^* \mid w \text{ has an even number of 1's}\}$ properly?

First, prove: *Every word accepted by $\mathcal{A}$ has an even number of 1's.*

Then, prove: *Every word with an even number of 1's is accepted by $\mathcal{A}$.*

Proof (cont'd)

By induction, we can prove that: for all $n \in \mathbb{N}$, the run of $\mathcal{A}$ on any word w of length n ends in the state "p # of 1's" where p is the parity of the number of 1's in w.

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Base case: $|w| = 0$, i.e., $w = \varepsilon$. Trivial.

Induction step: Assume it's true for words of length n. Consider a word $w = w_1 \dots w_n w_{n+1}$ of length $n+1$. Apply the induction hypothesis to $w_1 \dots w_n$ of length n and then consider the last letter w_{n+1}. So the run of $\mathcal{A}$ on $w_1 \dots w_n$ ends in the state "p # of 1's" where p is the parity of the number of 1's in $w_1 \dots w_n$. The run of $\mathcal{A}$ on w is one step longer than the run on the prefix $w_1 \dots w_n$. If $w_{n+1} = 0$, the end state remains the same and p is also the parity of the number of 1's in w;

Proof (cont'd)

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Proof (cont'd)

Every word accepted by $\mathcal{A}$ has an even number of 1's.

Proof (cont'd)

Every word accepted by $\mathcal{A}$ has an even number of 1's. Indeed, if $\mathcal{A}$ accepts word w, the run on it ends in the accepting state "even # of 1's". By our previous result, the parity of the number of 1's in w is then even.

Proof (cont'd)

Every word accepted by $\mathcal{A}$ has an even number of 1's. Indeed, if $\mathcal{A}$ accepts word w, the run on it ends in the accepting state "even # of 1's". By our previous result, the parity of the number of 1's in w is then even.

Every word with an even number of 1's is accepted by $\mathcal{A}$.

Proof (cont'd)

Every word accepted by $\mathcal{A}$ has an even number of 1's. Indeed, if $\mathcal{A}$ accepts word w, the run on it ends in the accepting state "even # of 1's". By our previous result, the parity of the number of 1's in w is then even.

Every word with an even number of 1's is accepted by $\mathcal{A}$. Indeed, if w has an even number of 1's, then, by our previous result, the run of $\mathcal{A}$ on w ends in the state "even # of 1's", which is an accepting state.

Venture a guess

